

Read as You See: Guiding Unimodal LLMs for Low-Resource Explainable Harmful Meme Detection

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Abstract

Detecting harmful memes is crucial for safeguarding the integrity and harmony of online environments, yet existing detection methods are often resource-intensive, inflexible, and lacking explainability, limiting their applicability in assisting real-world web content moderation. We propose **U-CoT+**, a resource-efficient framework that prioritizes accessibility, flexibility and transparency in harmful meme detection by fully harnessing the capabilities of lightweight unimodal large language models (LLMs). Instead of directly prompting or fine-tuning large multimodal models (LMMs) as black-box classifiers, we avoid immediate reasoning over complex visual inputs but decouple meme content recognition from meme harmfulness analysis through a high-fidelity meme-to-text pipeline, which collaborates lightweight LMMs and LLMs to convert multimodal memes into natural language descriptions that preserve critical visual information, thus enabling text-only LLMs to “see” memes by “reading”. Grounded in textual inputs, we further guide unimodal LLMs’ reasoning under zero-shot Chain-of-Thoughts (CoT) prompting with targeted, interpretable, context-aware, and easily obtained human-crafted guidelines, thus providing accountable step-by-step rationales, while enabling flexible and efficient adaptation to diverse sociocultural criteria of harmfulness. Extensive experiments on seven benchmark datasets show that U-CoT+ achieves performance comparable to resource-intensive baselines, highlighting its effectiveness and potential as a scalable, explainable, and low-resource solution to support harmful meme detection¹.

CCS Concepts

• **Computing methodologies** → **Discourse, dialogue and pragmatics; Reasoning about belief and knowledge.**

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¹Codes and data are available at: <https://github.com/panFJCharlotte98/HMC>



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Keywords

Harmful meme detection; Low-resource; Explainability; LLMs; LMMs

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1 Introduction

As a popular communication medium on social media platforms [24, 38], memes convey rich information in a visually engaging manner by combining images with overlaid text [13]. Despite their often amusing nature, memes can be exploited to disseminate harmful content under the guise of humor, such as hate speech [14, 26], toxic misinformation [25, 35], and inflammatory opinions [15, 32]. Harmful memes² pose a threat to the safety and harmony of online environments [42, 53], thus motivating automated harmful meme detection to support content moderation for online platforms.

However, existing methods, most of which follow a typical supervised fine-tuning (SFT) paradigm, face three critical limitations: **(i) Poor Resource Efficiency.** On one hand, supervised approaches tend to be data-inefficient as they require large volumes of labeled training data to perform well [7, 28, 45, 52]. However, the nuanced, implicit and context-specific nature of memes [5, 12] makes it difficult for human annotators to maintain consistent labeling without being limited by individual knowledge and subjectivity [18]. As such, collecting and annotating memes are often time-consuming, costly, and prone to bias. On the other hand, existing low-resource methods [21, 23, 30] often rely on proprietary Large Multimodal Models (LMMs) e.g., GPT-4 or heavily scaled models with massive parameter counts, incurring substantial financial and computational expenses. Taken together, these methods pose significant accessibility challenges for practitioners operating under low-resource constraints. **(ii) Limited Scalability and Flexibility.** The classification criteria distinguishing harmful and harmless memes often vary across sociocultural contexts [53, 71] and differ across platforms, regions, and over time [6, 19, 43, 54], which entails the

²Disclaimer: This paper contains demonstration content that may be disturbing.

use of specialized policies for different scenarios in order to robustly define harmful memes [58]. However, supervised classifiers are not easily scalable. Once trained, they become difficult to adapt to new out-of-domain criteria, meme or harmfulness types without undergoing costly data collection and retraining [11, 36]. (iii) **Lack of Explainability**. Real-world content moderation practices combine model-generated decisions to assist human verdicts [6, 16]. However, most supervised methods operate as black boxes, offering no human-interpretable reasoning behind their predictions [7, 8, 70]. This lack of transparency in decision-making undermines trust in human-machine collaboration and deployment reliability in sensitive or regulated scenarios [51], where explicit rationales are essential to support subsequent human reviews and feedback.

To address these limitations, in this paper, we explore the feasibility of harnessing lightweight Large Language Models (LLMs) for low-resource harmful meme detection. Specifically, we propose **U-CoT+**, a unimodal guided Chain-of-Thoughts (CoT) [61] prompting framework, which converts the default multimodal setup back to unimodal text-only inference, enabling LLMs to perform zero-shot harmful meme detection solely from textual input. To this end, we first propose a **High-Fidelity Meme2Text** pipeline. Instead of directly prompting LLMs to perform classification, we query lightweight LLMs multiple times to extract critical visual cues [9, 23] for sensitive, identity-related attributes of human subjects (e.g., race, gender). We then resort to a unimodal LLM to integrate the LLM outputs into a unified, informative description which it subsequently uses for detection. This design effectively decouples content recognition from harmfulness analysis, reducing task complexity by avoiding immediate reasoning over complicated visual elements.

Furthermore, we introduce the **Guided CoT** approach. Rather than relying solely on LLMs' internalized knowledge, we guide them with explicit, human-crafted guidelines that specify fine-grained, targeted criteria for distinguishing harmful from harmless memes. LLMs reason step by step over textual meme descriptions in conjunction with the guidelines, promoting consistent, interpretable, and aligned classifications. This also allows easy and effective adaptation to different harmfulness criteria across diverse sociocultural contexts with only minor resources and human effort.

Through extensive experiments on benchmark datasets, we show that our proposed zero-shot harmful meme detection framework effectively narrows the performance gap to state-of-the-art full-shot supervised fine-tuned baselines. Notably, with guided CoT prompting, lightweight LLMs (such as Mistral-12B and Qwen2.5-14B) perform competitively on zero-shot inference, matching and even outperforming GPT-4o(-mini) and other low-resource baselines. We summarize our main contributions as three-fold:

- We propose a harmful meme detection framework that effectively leverages the inherent image understanding abilities, commonsense knowledge and reasoning abilities of lightweight open-source LLMs and LLMs to enable explainable meme classification under low-resource settings. By performing zero-shot inference on high-fidelity meme descriptions under guided CoT prompting, our framework can easily adapt to different harmfulness evaluation criteria across diverse sociocultural contexts.
- We contribute a set of interpretable and targeted guidelines for harmful meme detection across various contexts. We demonstrate that these high-quality yet simple, human-crafted guidelines,

designed to capture diverse forms and features of harmfulness, can effectively enhance the zero-shot performance of LLMs in detecting harmful memes.

- We benchmark the zero-shot harmful meme detection performance of representative lightweight unimodal LLMs. We show that our method fully leverages their inherent capabilities, achieving performance comparable and even superior to that of resource-intensive multimodal methods. This highlights an accessible yet effective solution for multimodal tasks in low-resource scenarios.

2 Related Work

Harmful Meme Detection. This task seeks to classify online memes as harmful or harmless. A major line of works explore contrastive learning models and frameworks (e.g., based on CLIP [48]) to train vision language models as multimodal classifiers that fuse cross-modal features and learn unified meme representations in embedding space for image classification [7, 28, 36, 55, 59, 70]. Others generate image captions and fine-tune binary text classifiers based on pre-trained language models [10, 23, 56]. Some injects explainability in detection by prompting advanced LLMs [20, 30] or LLMs [29] to generate reasoning outputs for supervised training. However, existing work on low-resource harmful meme detection remains limited. The use of advanced multimodal models such as LLaVA-34B and GPT-4o as training-free detection agents is explored in [21]. Yet, small-scale LLMs struggle with zero-shot detection [1, 50].

LLMs with CoT Reasoning. LLMs extensively pretrained on universal knowledge have demonstrated remarkable zero-shot inference abilities over manifold reasoning tasks [27, 41, 46, 60, 62–68]. The Chain-of-Thought (CoT) method [61] aims to improve inference performance by prompting LLMs to explicitly generate intermediate reasoning steps before arriving at a final verdict. Recent studies have shown the effectiveness of CoT-enhanced supervised fine-tuning and LLMs (e.g., GPT-4V) for meme-centered tasks [29, 31, 69]. However, whether and how this applies to lightweight unimodal LLMs for harmful meme detection remains largely unexplored.

3 Methodology

The criteria for defining harmful content vary widely with sociocultural contexts and differ across platforms, regions and even over time. The use of diverse specialized policies in real-world content moderation highlights the need for context/domain-specific guidelines. By nature, harmful meme detection is a linguistic reasoning task that relies on the understanding of: (i) **Nuances in language use**, including metaphors, sarcasm, humor and other rhetorical devices; (ii) **Commonsense knowledge** such as social norms, moral values, sociopolitical factors and historical, cultural context; (iii) **Explicit criteria** that define the decision boundary between harmful and benign content. Unimodal text-only LLMs, pre-trained on large-scale web corpora and further aligned with human preferences, inherently possess extensive world knowledge and fundamental reasoning abilities. Trained with refined objectives specifically optimized for general-purpose reasoning and instruction following, they demonstrate more reliable performance on language-only tasks than multimodal counterparts of comparable size. Their availability at various scales also offers flexible options for low-resource deployment.

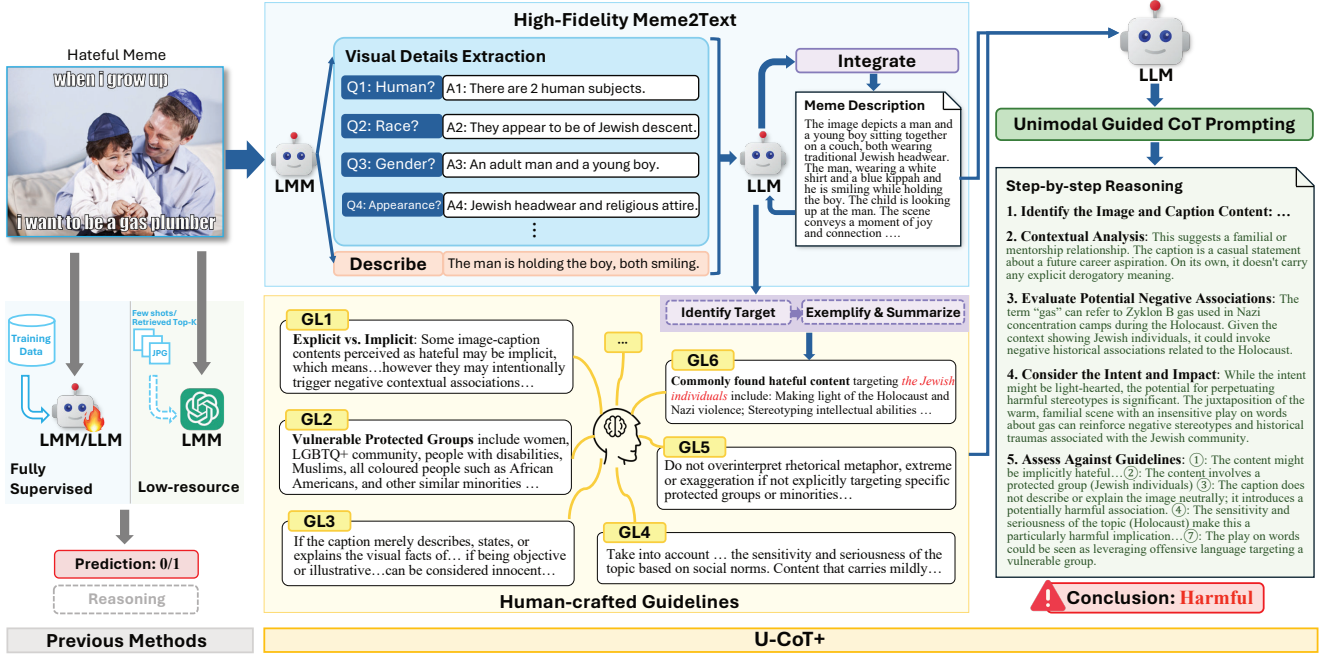


Figure 1: Illustration of previous methods and our U-CoT+. Previous methods follow either fully supervised or low-resource settings, fine-tuning LMMs/PLMs with labeled data or prompting advanced LMMs (e.g., GPT-4) with/without few-shot examples or retrieval-augmented mechanisms. They do not necessarily guarantee predictions that include explicit reasoning. U-CoT+ employs a High-fidelity Meme2Text pipeline to convert the multimodal harmful meme detection task into a unimodal, text-only setting, and further enhances LLMs’ reasoning through Unimodal Guided CoT Prompting. An example output given by Qwen2.5-14B under U-CoT+ is presented in “Step-by-step Reasoning”. The meme is only for demonstration purposes.

Therefore, we propose U-CoT+, a unimodal guided CoT framework that aims to harness the inherent abilities of lightweight LLMs in zero-shot inference for low-resource harmful meme detection. Adopting an empirically grounded, multi-stage LMM+LLM collaborative workflow [2, 3, 22], our approach effectively converts the default multimodal classification task into a unimodal, text-based linguistic reasoning setup, allowing LLMs to process memes based on textual descriptions and analyze meme harmfulness guided by human-crafted guidelines. The framework comprises two key components: (1) a **High-Fidelity Meme2Text** pipeline and (2) **Unimodal Guided CoT Prompting**, as demonstrated in Figure 1.

3.1 High-Fidelity Meme2Text

Typical harmful memes use brief, salient visual summaries to attack people’s identities. Interpreting their punchlines does not require exhaustive, irrelevant visual details, but rests on a clear understanding of what or who is being targeted. Commonly found vulnerable identity-related attributes critical for harmfulness detection include *race*, *gender*, *disability*, *physical appearance*, and *celebrity*, etc. However, safety-aligned LLMs, due to debiasing considerations in general-purpose applications, tend to omit socially and culturally sensitive visual cues but use ambiguous, identity-neutral references (e.g., “a person” or “some people”) when describing images with human subjects [23], which poses challenges for harmful content detection. To retain such essential content,

we decompose meme description generation into individual steps of visual information extraction via atomic question answering (VQA), which does not necessarily require highly capable models. Therefore, by repeatedly querying a lightweight LMM (such as LLaVA1.6-7B), each time focusing on a specific attribute, we can gather these key information without resorting to more resource-intensive LMMs. In this way, we effectively decouple meme content recognition from meme classification reasoning. This helps reduce task complexity by avoiding immediate reasoning over complicated, entangled visual elements, which is prone to hallucination.

Specifically, for each meme M , LMMs are given a general “Describe” prompt P^D such as “What is shown in the meme?” to generate a basic low-fidelity description: $D_l = \text{LMM}(M, P^D)$, and then identify the presence of any human subject. For memes showing humans, LMMs are further prompted to provide identify-related information $I_i = \text{LMM}(M, P_i^I)$, where i denotes different attributes. With all visual cues ready, we further leverage unimodal LLMs to integrate all LMM outputs into a unified, compact, and coherent high-fidelity description of the meme content: $D_h = \text{LLM}(D_l, I_r, I_g, I_d, I_a, I_c)$, which serves as input for the subsequent text-based classification.

3.2 Unimodal Guided CoT Prompting

3.2.1 Guideline Crafting. The subjective and context-dependent nature of harmful meme definitions makes it unlikely to eliminate

human involvement in guideline crafting process. However, harmful meme detection largely relies on widely accessible commonsense knowledge rather than specialized expert insights. Accordingly, we propose an empirical guideline crafting procedure that incurs only minor human effort: (i) We examine the available harmful meme definitions and data annotation guidelines in existing datasets for the patterns of harmfulness in representative memes; (ii) We summarize the observed patterns and underlying criteria into coherent natural language guidelines consisting of multiple rules ordered in a logically smooth way (e.g., from general to specific, from features to examples, etc.); (iii) We refine guidelines by probing their clarity and effectiveness on sample memes. This procedure yields the following set of principles for guideline formulation that are reproducible and transferable across different sociocultural contexts:

- **Implicitness** Harmful memes can be implicit [21]. This principle requires LLMs to reflect on the underlying implications of a meme to examine whether it is deliberately crafted to evoke negative contextual interpretations (e.g., associations to harmful stereotypes, sensitive controversies in societal, cultural or political practices etc.) while wrapping harmful messages beneath deceptively benign visuals and neutral language.
- **Tone & Intent** LLMs tend to interpret memes charitably due to alignment with human preferences for safe responses. To mitigate such tendency, this principle explicitly instructs LLMs to avoid assuming a playful, humorous or light-hearted tone and intent, and instead interpret from a neutral perspective.
- **Fine-grained Taxonomy** Context-specific harmful memes often have fine-grained taxonomies of *target* or *type*. A harmful meme’s **target** is the entity toward which its harmful message is directed, such as a vulnerable minority, protected group, organization or specific public figure, etc. The **type** of a harmful meme refers to the sub-category of harmfulness it falls in, such as “*stereotypes*” or “*body shaming*” etc., in misogynistic memes [17]. Based on such taxonomies, by first prompting LLMs to identify a meme’s potential target or breaking down into fine-grained classification sub-tasks, this principle seeks to narrow the problem and reduce uncertainty, guiding more focused reasoning through the use of target- or type-specific guidelines.
- **Patterns** This principle aims to foster analogical reasoning by prompting LLMs with distilled patterns of harmful memes e.g., “*Perpetuating stereotypes on female’s domestic roles*”, “*Anti-Semitic contents that make light of the Holocaust*” etc. To derive such patterns with further reduced human efforts, we can leverage LLMs’ in-context learning abilities to extend and integrate human-provided examples into structured heuristics.
- **Exception** When detecting harmful memes, it is crucial not to prioritize high recall at the cost of precision. This principle aims to prevent over-censorship by introducing exception cases in which memes should be considered “harmless” despite containing traces of potentially sensitive content. These exceptions may vary across contexts and domains, often reflecting specific evaluation criteria or inherent biases in data annotation.

3.2.2 Zero-shot CoT Prompting. Following the widely adopted implementation of chain-of-thought (CoT) reasoning [61], we prompt unimodal LLMs in a **zero-shot** setting using a heuristic prompt P^{CoT} such as “*Now, let’s think step by step*” or “*Now, let’s apply the*

guidelines one by one”. This encourages LLMs to produce the final classification decision CLS through detailed step-by-step rationales R by reasoning over the high-fidelity meme description D_h together with our human-crafted harmful meme detection guidelines GL . Consequently, we have: $CLS, R = \text{LLM}_{U-CoT+}(D_h, GL, P^{CoT})$.

4 Experiments

We report the Accuracy (**Acc.**) and macro-F1 (**F1**) score for performance evaluation in line with recent studies [21, 36]. As our method extracts LLM predictions based on exact token match, probability-based AUC scores are not applicable. We report the corresponding accuracy results for baselines that evaluate using AUC. See Section B for implementation details.

Datasets. We evaluate our proposed framework on seven widely used harmful meme detection datasets: (i) **FHM**[26], (ii) **HarMeme** [44], (iii) **Harm-P**[45], (iv) **MultiOFF**[57], (v) **MAMI**[17], (vi) **PrideMM**[52] and (vii) **GoatBench**[31]. These datasets and benchmarks cover various types of harmfulness against diverse targets across a wide range of sociocultural contexts.

Baselines. For **full-shot SFT** methods, we consider: (i) LMM-RGCL [36], (ii) UMR[70], (iii) ExplainHM[30], (iv) Pro-Cap[9], (v) IntMeme [20], (vi) CapAlign[23], (vii) M3Hop-CoT[29], (viii) MemeCLIP[52]. For **low-resource training-free** methods, we include (i) Zero-shot inference with GPT-4o-mini[40], (ii) LoREHM[21], (iii) MIND[34], (iv) Mod-HATE[11], and (v) GPT-4V[39].

Models. We experiment with representative state-of-the-art lightweight open-source models, including LLaVA1.6-7B [33] and Qwen2 VL-7B[47] as LMMs, and instruction-tuned Qwen2.5-14B and 7B [49], Mistral-12B [37], and Llama3.1-8B [4] as unimodal LLMs.

Table 1 and Table 2 compare our best zero-shot harmful meme detection performance achieved by the best-performing unimodal LLMs under U-CoT+ to all the state-of-the-art (SOTA) baselines under either full-shot SFT or low-resource settings. We denote the traditional LMM-based multimodal classification scheme under zero-shot CoT as **M-CoT**, where the LMM reasons over the prompt and image. **U-CoT** denotes **U-CoT+**’s ablation variant, where the LLM performs zero-shot CoT reasoning over textual inputs but without referring any supportive information such as guidelines.

4.1 Comparing U-CoT+ to Baselines

U-CoT+ vs. Supervised Fine-tuned Baselines As shown in Table 1, performance gaps remain between the state-of-the-art fully supervised fine-tuned baselines and lightweight LLMs under our proposed unimodal guided zero-shot CoT prompting, but they have narrowed to a generally acceptable and promising range, especially given our focus on low-resource settings. Specifically, LLMs’ zero-shot performance has become nearly comparable to that of full-shot baselines on FHM, HarMeme, MAMI and PrideMM. On FHM, Mistral-12B outperforms IntMeme and comes within 3 points of the performance of ProCap and ExplainHM. Notably, its performance on HarMeme is highly comparable to that of most SFT baselines, surpassing IntMeme and CapAlign, even with better macro-F1. Moreover, both LLMs perform well on MAMI, with Qwen2.5-14B matching the performance of LMM-RGCL. Its best zero-shot result on the challenging PrideMM also falls within 7 points to that of the two SFT baselines. On the U.S. political meme dataset Harm-P,

Table 1: Comparison with SOTA SFT and low-resource baselines. The best SFT results are in bold. The best zero-/few-shot performance is highlighted in bold and underlined; second-best is underlined. †: RGCL+RKC/FS is the out-of-distribution performance of LMM-RGCL [36] with retrieval augmentation and few-shot examples, respectively.

Dataset		FHM		HarMeme		Harm-P		MultiOFF		MAMI		PrideMM	
Method	Base Model	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1
<i>Full-shot Supervised Fine-tuned Classifiers</i>													
LMM-RGCL	Qwen2VL-7B	82.10	–	88.10	–	91.60	91.10	71.10	64.80	79.90	–	78.10	78.40
UMR	BLIP	77.20	–	87.85	86.99	92.11	92.11	–	69.96	–	–	–	–
ExplainHM	GPT3.5+FLAN-T5	75.60	75.39	87.00	86.41	90.73	90.72	–	–	–	–	–	–
Pro-Cap	BLIP-2+RoBERTa	75.10	–	85.03	–	–	–	–	–	73.63	–	–	–
IntMeme	FLAVA+RoBERTa	71.52	–	82.66	–	–	–	–	–	72.44	–	–	–
CapAlign	ChatGPT+FLAN-T5	–	–	82.20	68.63	93.80	93.11	–	–	–	–	–	–
M3Hop-CoT	CLIP etc.+Mistral-7B	–	–	–	–	–	–	–	–	80.28	–	–	–
MemeCLIP	CLIP	–	–	–	–	–	–	–	–	–	–	76.06	75.09
<i>Low-resource Baselines w/ Retrieval Augmented Classification</i>													
LoReHM	LLaVA-34B	65.60	65.59	73.73	70.86	–	–	–	–	75.40	75.28	–	–
LoReHM	GPT-4o	70.20	70.14	74.54	72.98	–	–	–	–	83.00	82.98	–	–
MIND	LLaVA1.5-13B	60.80	60.71	68.93	65.19	–	–	–	–	68.90	68.84	–	–
RGCL+RKC†	Qwen2VL-7B	69.30	–	81.90	–	67.30	67.80	63.80	55.60	75.60	–	69.30	69.30
<i>Low-resource Baselines w/o Retrieval Augmented Classification</i>													
Zero-shot	GPT-4o-mini	67.60	65.51	70.90	69.46	<u>65.35</u>	<u>65.35</u>	<u>65.77</u>	<u>64.86</u>	77.40	76.59	72.39	72.28
RGCL+FS†	Qwen2VL-7B	63.50	–	68.10	–	–	–	–	–	–	–	–	–
Mod-HATE	BLIP-2+LLaMA-7B	57.60	53.88	71.19	69.64	–	–	–	–	69.05	68.78	–	–
<i>Ours: Unimodal Guided Zero-shot CoT</i>													
U-CoT+	LMM _{7B} +Qwen2.5-14B	72.50	72.41	83.62	82.00	65.35	65.35	69.13	68.37	79.90	79.89	71.60	71.37
	LMM _{7B} +Mistral-12B	72.90	72.87	83.90	82.41	63.10	62.01	64.43	61.36	75.00	74.72	69.03	68.07

Table 2: Performance on Goat-Bench. S.: Scheme. M: M-CoT; U: U-CoT; U+: U-CoT+. GPT-4V results are reported in [31]. Our best results are highlighted in bold.

Model	S.	Hateful		Harmful		Misogyny		Offensive	
		Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1
GPT-4V	M	71.90	71.37	66.90	64.97	81.60	81.57	61.37	60.99
Qwen2.5-14B	U	70.80	69.17	62.64	60.51	77.40	77.38	60.97	59.76
	U+	71.95	71.05	71.16	70.69	80.10	80.09	61.78	60.90
Mistral-12B	U	70.20	68.21	61.45	58.93	75.50	75.48	63.53	60.62
	U+	71.45	70.43	70.76	68.32	75.10	74.83	60.70	58.86

SFT methods consistently achieve high accuracy over 90%, whereas zero-shot inference, even with GPT-4o-mini, struggles to exceed 70%, which reflects the complex, highly subjective, heterogeneous, and target-dependent nature of defining harmfulness in Harm-P. In contrast, SFT methods perform less strongly on MultiOFF, despite its contextual and stylistic similarity to Harm-P, making Qwen2.5-14B’s zero-shot performance comparable. SFT’s decline on MultiOFF suggests a drawback of reduced training data (455 vs. 2.9K in Harm-P). This observation verifies SFT methods’ heavy reliance on large-scale training data, revealing their inherent limitations under low-resource conditions.

U-CoT+ vs. Low-Resource Baselines Low-resource baselines with retrieval-based augmentation such as LoReHM, MIND, though operating under a zero-shot setting, further benefit from a majority voting of ground-truths of the top-K most similar memes retrieved from the embedding space. In contrast, U-CoT+ operates under a more constrained low-resource setting, without additional access to auxiliary ground-truth signals from neighbour memes as reference for decision makings. Despite the absence of retrieval augmentation, we show that lightweight LLMs can unlock their full potential for zero-shot harmful meme detection. As shown in Table 1, with the support of our human-crafted guidelines, both Qwen2.5-14B and Mistral-12B can largely outperform MIND, and can outperform the other two retrieval-augmented baselines on FHM and HarMeme, even surpassing LoReHM’s performance based on GPT-4o. Qwen2.5-14B can further outperform LMM-RGCL+RKC on MultiOFF, MAMI and PrideMM. Notably, the diminished out-of-distribution performance of LMM-RGCL, even when enhanced by the retrieval mechanism (RKC), further demonstrates the limited generalizability of SFT methods. Furthermore, U-CoT+ can also significantly outperform LMM-RGCL+FS and Mod-HATE, both of which are baselines without retrieval augmentation but are evaluated under few-shot settings. Besides, lightweight LLMs under U-CoT+ consistently outperform or match GPT-4o-mini’s zero-shot performance based on multimodal inference across five of six datasets, with slightly weaker results only observed on PrideMM. Table 2 compares the zero-shot performance of our method to that

Table 3: Ablation studies. The best low-resource results (with open-source models, without ensemble) are in bold, with the second-best underlined. Δ : Performance gain over 0-shot guideline-free U-CoT. $\bar{\Delta}_{U-CoT+}$: The average performance gain of U-CoT+ under guideline perturbations. $\Delta_{U-CoT+L}$: The performance gain of U-CoT+ with LoREHM insights. $\bar{\Delta}_{U-CoT+FS}$: The average performance gain of U-CoT+ with few-shot (4,6,8 and 10-shot) examples.

LLM	LMM	Scheme	FHM		HarMeme		Harm-P		MultiOFF		MAMI		PrideMM	
			Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1
N/A	GPT-4o-mini	–	67.60	65.51	70.90	69.46	65.35	65.35	65.77	64.86	77.40	76.59	72.39	72.28
	Qwen2VL-7B LLaVA1.6-7B	M-CoT	64.20 60.40	62.68 57.85	67.51 66.38	60.29 61.05	56.34 56.34	53.05 53.62	71.14 59.73	64.61 57.38	68.10 67.80	66.03 67.46	68.44 60.16	68.43 59.99
Qwen2.5-14B	7B LMM	U-CoT	<u>70.10</u>	<u>70.02</u>	59.60	51.46	<u>60.00</u>	<u>59.91</u>	64.43	62.93	<u>76.40</u>	<u>76.38</u>	66.47	66.47
		U-CoT+	<u>72.50</u>	<u>72.41</u>	<u>83.62</u>	<u>82.00</u>	65.35	65.35	69.13	68.37	79.90	79.89	71.60	71.37
		$\bar{\Delta}_{U-CoT+}$	1.78	1.75	22.11	28.58	4.22	4.23	0.50	0.98	1.96	1.93	3.70	3.60
		$\Delta_{U-CoT+L}$	0.40	0.47	7.63	14.44	–	–	–	–	0.30	0.32	–	–
		$\bar{\Delta}_{U-CoT+FS}$	-0.70	-0.62	4.81	1.78	1.20	1.21	-0.67	-1.77	1.25	1.25	0.44	0.26
	7B LMMs GPT-4o-mini	Ensemble U-CoT+	72.40 73.00	72.23 73.00	82.77 83.90	82.13 83.05	65.07 62.25	65.05 62.25	62.42 60.40	62.21 60.22	79.90 79.10	79.85 78.78	↑73.37 68.44	↑73.36 68.39
Mistral-12B	7B LMM	U-CoT	<u>67.20</u>	<u>66.70</u>	65.54	<u>63.48</u>	<u>58.87</u>	<u>58.32</u>	61.74	56.89	<u>73.00</u>	<u>72.99</u>	62.72	62.16
		U-CoT+	72.90	72.87	83.90	82.41	<u>63.10</u>	<u>62.01</u>	64.43	61.36	74.90	74.81	<u>69.03</u>	<u>68.07</u>
		$\bar{\Delta}_{U-CoT+}$	4.22	4.70	14.76	14.94	4.51	4.04	2.69	4.43	2.20	2.09	6.46	6.49
		$\Delta_{U-CoT+L}$	-1.40	-2.92	0.00	-4.33	–	–	–	–	-5.50	-7.75	–	–
		$\bar{\Delta}_{U-CoT+FS}$	-0.73	-0.96	1.20	-4.78	1.77	-1.74	-0.67	-5.70	1.15	0.77	0.79	-1.41
	7B LMMs GPT-4o-mini	Ensemble U-CoT+	↑73.50 71.10	↑73.43 71.09	83.90 81.92	82.86 80.61	63.10 62.82	62.16 61.27	63.09 67.79	61.53 66.12	↑76.00 75.20	↑75.61 74.84	↑70.02 66.27	↑70.02 66.21
Llama3.1-8B	7B LMM	U-CoT	63.50	61.44	62.43	50.64	<u>62.25</u>	<u>58.78</u>	63.76	57.55	<u>68.70</u>	<u>68.10</u>	60.55	58.49
		U-CoT+	<u>68.00</u>	<u>66.98</u>	<u>71.19</u>	<u>61.44</u>	<u>61.13</u>	<u>56.21</u>	68.46	62.35	<u>73.50</u>	<u>73.50</u>	64.50	63.26
	7B LMMs	Ensemble	↑69.50	↑69.00	↑75.14	↑69.34	60.56	56.29	67.79	61.79	↑74.00	↑74.00	↑67.85	↑67.58
Qwen2.5-7B	7B LMM	U-CoT	<u>66.70</u>	<u>66.45</u>	59.89	59.82	<u>58.59</u>	<u>57.46</u>	63.76	63.72	<u>70.60</u>	<u>70.59</u>	63.51	62.38
		U-CoT+	<u>67.50</u>	<u>67.50</u>	<u>77.12</u>	<u>75.26</u>	<u>62.54</u>	<u>60.20</u>	<u>67.11</u>	<u>66.63</u>	<u>74.40</u>	<u>74.40</u>	67.26	66.90
	7B LMMs	Ensemble	67.00	66.98	70.34	70.23	↑62.82	↑61.76	63.09	63.09	↑75.80	↑75.78	67.06	67.03

Note: U-CoT results outperforming M-CoT are blue. U-CoT+ results based on ≤8B LLMs that outperform M-CoT are green. ↑: Ensemble results further improving on U-CoT+.

of GPT-4V on Goat-Bench [31]. As can be seen, U-CoT+ enables lightweight unimodal LLMs to achieve performance highly comparable to, and in some cases even surpassing that of the advanced multimodal GPT-4V across all four sub-tasks.

In summary, results validate U-CoT+’s advantages: (i) It effectively leverages lightweight LMMs and LLMs, and achieves highly comparable zero-shot performance relative to more resource-intensive baselines, thereby demonstrating strong resource efficiency. (ii) Its underlying pipeline and principles support empirical, context-specific adaptation to diverse detection criteria with only minor human efforts, offering flexibility. (iii) Its consistently top-ranked performance among low-resource methods across diverse datasets underscores the scalability of our guideline crafting process, which generalizes well across varied sociocultural contexts.

4.2 Ablation Study

Table 3 presents the main results of ablation studies that compare different prompting schemes and validate the effectiveness of key framework components. Figure 2 presents the detailed confusion matrices by Qwen2.5-14B.

Unimodal vs. Multimodal. We observe consistently improved zero-shot performance on FHM, Harm-P, and MAMI merely by converting the default multimodal setting to unimodal, text-based inference. This highlights the effectiveness of modality conversion, which enables certain meme messages to be more explicitly revealed

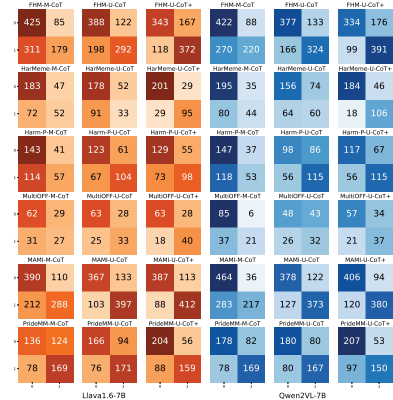


Figure 2: Confusion matrices of Qwen2.5-14B based on meme descriptions sourced from different 7B LMMs.

when conveyed in text. Figure 2 shows that both 7B LMMs exhibit an inherent bias toward the negative class, frequently overclassifying memes as “harmless” in most datasets. This evidences their limitation in detecting harmfulness based on visual inputs. Reasoning with more capable unimodal LLMs can, in most cases, help balance classification with more memes revealed to be “harmful”.

Effectiveness & Robustness. Comparing U-CoT+ against U-CoT results demonstrates the overall effectiveness of our human-crafted guidelines. Notably, U-CoT+ enables LLMs of comparable size to 7B LMMs to outperform M-CoT results across four datasets. We further apply different perturbation strategies *e.g.*, *Rephrasing*, *Shuffling*, and *Modifying* to our guidelines and compute the resulting average performance gain of U-CoT+ over U-CoT. As shown by $\bar{\Delta}_{U-CoT+}$ in Table 3, despite the potential randomness, semantic nuances and logic flow disruption introduced, our guidelines remain conceptually constructive, consistently improving performance across most datasets. This also evidences U-CoT+’s general robustness to guideline perturbations.

Human-crafted vs. GPT-generated. We compare the effectiveness of our human-crafted guidelines with GPT-4o-generated harmful meme detection insights from LoReHM [21]. As shown in Table 3 ($\bar{\Delta}_{U-CoT+}$ vs. $\Delta_{U-CoT+L}$), prompting LLMs with human-crafted guidelines in U-CoT+ yields greater improvements over zero-shot U-CoT than using GPT-generated insights, which may even lead to performance drops when applied to Mistral-12B. These GPT-4o-summarized insights alone, which tend to be overly general and verbose, lacking explicit context-specific relevance and rule-level diversity, prove insufficient for effectively guiding lightweight unimodal LLMs in harmful meme detection. This further underscores the usability and effectiveness of our guideline crafting principles in providing targeted, interpretable guidance that is easy to follow.

Guidelines vs. Few shots. We evaluate four few-shot (FS) settings (4, 6, 8 and 10-shot) with class-balanced meme descriptions randomly sampled from the holdout pool (see Section C.2 for details of demonstration example construction.), and compare their average performance gain over zero-shot U-CoT ($\bar{\Delta}_{U-CoT+FS}$) to $\bar{\Delta}_{U-CoT+}$ in Table 3. Results show that, despite occasional improvements, providing LLMs with meme descriptions as few-shot demonstration examples is generally less effective than guideline-based prompting, and may even result in deteriorated performance compared to that of zero-shot inferences, suggesting that LLMs’ in-context learning ability does not translate well to this task, which aligns with previous findings for LMMs [21, 36].

7B LMMs vs. GPT-4o-mini for Meme2Text. We replace meme descriptions produced by our proposed pipelines with GPT-4o-mini-generated descriptions, which generically provide more accurate visual information. Results for U-CoT+ in Table 3 show that reasoning over meme descriptions generated by the more capable GPT-4o-mini does not consistently outperform those produced by less advanced lighter-weight LMMs, suggesting that the performance bottleneck is not primarily visual. This further indicates that our proposed meme2text pipeline, despite relying on low-resource models, is still effective in capturing visual information essential for harmfulness assessment.

Single vs. Ensemble. We compare single-LMM U-CoT+ with ensemble strategies that aggregate predictions from other scheme variants (*i.e.*, M-CoT and U-CoT), as well as from U-CoT+ based on different LMMs (See Table 6). Results in Table 3 show that even without relying only on a single best-performing 7B LMM, we can largely preserve performance through appropriate ensembles of predictions, which may even yield further improved results on MAMI and PrideMM. In practice, ensemble strategies help mitigate uncertainty in LLMs’ sensitivity to the choice of 7B LMMs for

Meme Example	(a)	(b)
w/o Guidelines	The image shows three Middle Eastern men standing outdoors in an urban setting with modern buildings. They are dressed traditionally in white robes and head coverings. All of them are pointing upwards, suggesting they might be looking at something in the sky. "Your uncle Samir" is likely a reference to someone known to the viewer, perhaps making light of the idea that this plane belongs to their relative named Samir... The tone appears to be playful rather than malicious or hateful. While the caption is casual and potentially stereotypical (implying that "Samir" might own a plane), it does not appear to be meant to offend or harm anyone...no derogatory language...it seems to be a light-hearted or humorous comment... Conclusion: The content can be classified as non-hateful (X)	The image appears to be a form of political commentary or satire, using humor and exaggeration to criticize Hillary Clinton's policies and positions...suggests a sarcastic and mocking tone. The statements listed are indeed controversial. Some touch upon serious moral and political sensitivities... However, this meme seems to lean more towards humor than harassment given its clear satirical tone...some viewers might find the content offensive, especially if they strongly support Clinton or believe the statements are unfairly exaggerated...In the context of social media...such memes are not uncommon and are generally accepted as part of the broader conversation... Conclusion: common practices of political satire non-offensive (X) . However, political sensitivity varies with individuals. What one person finds acceptable may offend another.
w/ Guidelines	Following the provided guidelines step-by-step...The phrase "was it your uncle Samir" is a play on words that attempts to make a joke by implying a connection between the men's actions and a stereotype that associates Middle Eastern individuals with terrorism or extremist ideologies, a particular event (possibly referring to the September 11 attacks), which is highly problematic and offensive...GL 2: The content involves stereotypes against a protected group...GL 3: The caption is not simply describing the image; it includes a loaded statement that evokes negative contextual associations. GL 4...a stereotype that has historically been used to discriminate and spread hate towards Middle Eastern individuals... Conclusion: Its implication reinforces harmful biases...should be classified as hateful (V)	Apply the provided guidelines carefully: GL2: Some statements might not accurately represent her views or actions...claims are taken out of context...attributing exaggerated controversial positions to her could spread misinformation...perpetuates a negative stereotype about her, suggesting she supports harmful or unpopular policies...without providing balanced or nuanced information. GL 3: The overall tone is dismissive and aggressive...aims to ridicule, humiliate and discredit her...uses her name and image to link her with a series of unflattering and politically contentious statements...can be seen as a form of personal attack. GL4...clearly intended to appeal to those who oppose her or view her negatively...fosters partisan distrust and hostility... Conclusion: The image should be classified as offensive (V) accordingly.

Figure 3: Comparing LLM rationales before and after applying context-specific guidelines. Incorrect . Correct .

visual information extraction, while also alleviating potential over-censorship introduced by strict guideline-based assessment.

5 Case Study

Although grounded in a low-resource setup, U-CoT+ fully leverages the inherent knowledge and reasoning capabilities of lightweight LLMs for explainable harmful meme detection by prompting them to generate explicit step-by-step rationales leading to the final predictions. Comparing LLMs’ reasoning trajectories enables a closer examination of how our human-crafted guidelines influence their “mindset”, as illustrated in Figure 3. First, the guidelines provide LLMs with more aligned and context-specific guidance that enhances their associations, particularly for those negative interpretations with implicit harmful cues. For example, in Figure 3(a), when no guideline is provided, despite correctly recognizing the visual content, the LLM misinterprets the mention of “plane” as “owning a plane”, linking it to a neutral stereotype that assumes Middle Eastern individuals as wealthy. In contrast, when guidelines are presented, the LLM correctly identifies the sensitivity of the “plane” reference and associates it with a more serious implication of terrorism by pointing out its possible allusion to the 9/11 attacks, which is crucial for determining the meme as harmful. Second, the guidelines clarify the boundary between black and white by defining detailed checklists, thereby reducing uncertainty in the “gray zone”. For the meme shown in Figure 3(b), when guidelines are absent, the LLM reasonably acknowledges the meme’s potential offensiveness but ultimately downplays it as common political satire. With the guidelines in place, however, the LLM becomes more confident in

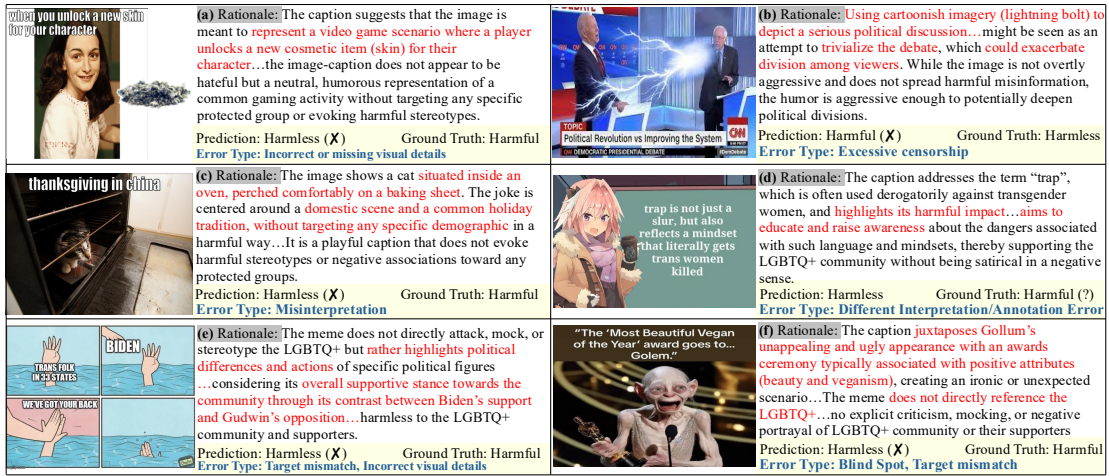


Figure 4: Examples of incorrectly classified memes and their corresponding error type.

affirming the meme as offensive based on matches with multiple criteria, such as misinformation, stereotyping, and personal attacks *etc.* As such, U-CoT+ enables human content moderators to intuitively formulate and apply detection guidelines, and to further refine them or verify models' verdicts based on reasoning outputs, thereby supporting more consistent and accountable decisions.

6 Error Analysis

U-CoT+'s transparency in decision makings enables fine-grained error analysis that reveals the inherent behaviors of lightweight LLMs (and LMMs) in harmful meme detection. We summarize common causes of misclassification by closely examining LLMs' rationales. Figure 4 presents typical misclassified examples and Figure 6 shows the quantitative distribution of different error types: **(i) Incorrect or missing visual details:** The image understanding and text recognition capabilities of 7B LMMs remain limited. For instance, the LMM fails to recognize Anne Frank in Figure 4(a), thereby missing the connection between the visual element "ash" and the insensitive caption. Moreover, LMMs still struggle to provide accurate details for visually noisy, stylistically abstract, blurred, or text-heavy memes, such as edited screenshots of chats or tweets. **(ii) Excessive censorship:** LLMs tend to overinterpret or overstate potential negative effects when they apply the guidelines too rigidly or adopt an overly strict perspective, such as interpreting sarcasm or parody in political debates shown in Figure 4(b) as trivializing serious issues. **(iii) Misinterpretation:** LLMs may misinterpret a meme's tone or intent when the meme is subtle, implicit, ambiguous, or poorly framed. Discrepancies between LLMs' and human interpretations of the guidelines can also cause LLMs to overlook or inconsistently apply them. **(iv) Different interpretations:** LLMs may arrive at reasonable interpretations that diverge from human annotations, largely due to differences in perceived levels of harm. In such cases, LLM judgments are not necessarily incorrect. For instance, LLMs may classify some annotated harmful memes as harmless by interpreting them as mild ridicule or satire that is common and generally acceptable in political discourse. These misclassifications may instead reflect inconsistent or erroneous annotations.

(v) Target mismatch: The inferred target subject of a meme determines which subset of guidelines is applied. Discrepancy in target identification can lead to guideline mismatches and subsequent misclassifications. Figure 4(e) is labeled as harmful toward President Biden, but the LLM instead interprets the target as the LGBTQ+ community and thus predicts the meme as harmless. **(vi) Blind Spot:** LLMs adhere to guideline-bounded reasoning and do not actively extend beyond it to account for long-tailed harmfulness types outside the main context. In Figure 4(f), the LLM fails to generalize to the vegan group under the PrideMM guidelines, which primarily focus on LGBTQ+ issues. These errors highlight limitations in both models and datasets, motivating future improvements: **(i)** Enabling general LLMs to rapidly learn and adapt to evolving popular culture and communication practices; and **(ii)** Developing multi-level, fine-grained, caption- and explanation-aware annotation frameworks for high-quality harmful meme detection benchmarks.

7 Conclusion

In this paper, we introduce U-CoT+, a novel framework for low-resource harmful meme detection. With a high-fidelity memetext pipeline, it decouples meme content recognition from harmfulness analysis and converts the default multimodal task into a unimodal, text-based setting, thus enabling detection based on LLMs' inherent capabilities. By incorporating targeted, interpretable and easily obtained human-crafted guidelines, U-CoT+ further enhances the zero-shot performance of lightweight LLMs with guided CoT prompting, achieving performance highly competitive with, or even surpassing that of resource-intensive baselines across datasets from diverse sociocultural contexts. Experiment results demonstrate U-CoT+'s effectiveness in offering strong resource efficiency, flexibility, scalability and explainability for harmful meme detection.

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A ETHICS STATEMENT

This work aims to fully harness the inherent capabilities of state-of-the-art LLMs for harmful meme detection under low-resource constraints, contributing to the safety and integrity of online communication, particularly on social media platforms. We position

our method as an early-stage support tool for real-world content moderation, designed to assist human moderators by providing LLM-generated verdicts as auxiliary references for human verification. In doing so, our method helps alleviate moderators’ emotional and cognitive burden. We see our work as part of the broader effort to foster a safe, respectful and responsible digital environment.

We exclusively use publicly available datasets in compliance with their usage agreements and release our codebase along with detailed implementation specifics to support reproducibility. Our work aligns with the vision of sustainable and accessible AI by leveraging lightweight open-source models that are deployable under common computational conditions.

While our guidelines are not inherently designed to induce bias, their effectiveness may be constrained by pre-existing biases embedded in the underlying LLMs. To mitigate the risk of unfair moderation arising from such LLM-induced biases, it is crucial that human moderators interpret LLM verdicts with caution. Our method is intended solely for the detection and prevention of harmful memes; any use that promotes, condones, or encourages hate speech or harmful content is strictly prohibited and strongly condemned.

B IMPLEMENTATION DETAILS

We conduct experiments on two NVIDIA A6000 48GB GPUs. All LMM-based inference are conducted with a batch size of 1. Qwen2.5 and Mistral are run with a batch size of 16. Llama3.1 is run with a batch size of 32. We obtain GPT-4o-mini inference results using the OpenAI API with the model version “gpt-4o-mini-2024-07-18”. For reproducibility, we adhere to greedy decoding without sampling (with “do_sample” set to False and “temperature” to 0) to ensure fully deterministic generation. The “max_new_tokens” is set as 256 or 356 for LMMs and 1024 or 1536 for LLMs depending on datasets. Table 5 shows detailed dataset statistics.

All prompt templates and workflows for visual information extraction, information integration, fine-grained target identification (e.g., protected group detection and hateful forms generation for FHM, entity identification and target classification for PrideMM) and guided CoT prompting are released in our codebase. Figure 5 visualizes how our proposed meme2text pipeline configures visual questions for different sociocultural contexts.

For ablation study, we query GPT-4o-mini using fine-grained prompts to obtain detail-preserving meme descriptions. For memes that GPT-4o-mini refuses to process (e.g., 14 in FHM and 332 in MAMI), we fallback to our original descriptions based on 7B LMMs. Table 6 illustrates our ensemble strategies that integrate predictions based on different LMMs as well as from U-CoT and M-CoT. Specifically, by default, if there is at least one positive U-CoT+ prediction, we classify the meme as harmful. For datasets where we have high confidence in the alignment between our guidelines and the true positives, memes with no positive U-CoT+ predictions are classified as harmless, regardless of the predictions made by M-CoT or U-CoT. For guidelines with a medium confidence level, memes are also considered harmless if there is only one positive U-CoT+ prediction while all M-CoT and U-CoT predictions are negative. When the guidelines are prone to over-censorship (i.e., with a low confidence level), memes with all negative M-CoT and U-CoT predictions are treated as harmless, regardless of the guideline-based results.

Table 4: Full experiment results. Δ : Performance gain over zero-shot guideline-free U-CoT.

LLM	LMM	Scheme	FHM		HarMeme		Harm-P		MultiOFF		MAMI		PrideMM	
			Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1
N/A	GPT-4o-mini	–	67.60	65.51	70.90	69.46	65.35	65.35	65.77	64.86	77.40	76.59	72.39	72.28
	Qwen2VL-7B	M-CoT	64.20	62.68	67.51	60.29	56.34	53.05	71.14	64.61	68.10	66.03	68.44	68.43
	LLaVa1.6-7B		60.40	57.85	66.38	61.05	56.34	53.62	59.73	57.38	67.80	67.46	60.16	59.99
Qwen2.5-72B	LLaVa1.6-13B		63.60	62.91	60.17	57.97	56.90	56.45	54.36	53.61	72.90	72.90	56.02	55.26
	Qwen2VL-7B	U-CoT	70.10	70.02	61.02	57.92	60.00	59.91	53.69	53.15	75.10	75.10	68.44	68.42
		U-CoT+	72.50	72.41	81.92	81.00	65.35	65.35	63.09	62.41	76.10	75.74	70.41	70.04
		Δ_{U-CoT+}	1.78	1.75	21.11	23.32	4.22	4.23	10.24	10.16	2.03	1.86	2.27	2.07
		$\Delta_{U-CoT+FS}$	-0.70	-0.62	6.35	6.17	1.20	1.21	5.37	4.27	2.70	2.70	1.28	1.02
	LLaVa1.6-7B	U-CoT	68.00	67.70	59.60	51.46	63.94	63.84	64.43	62.93	76.40	76.38	66.47	66.47
		U-CoT+	71.50	71.48	83.62	82.00	63.94	63.67	69.13	68.37	76.90	76.77	71.60	71.37
		Δ_{U-CoT+}	3.22	3.51	22.11	28.58	-1.33	-1.70	0.50	0.98	1.57	1.53	3.70	3.60
		$\Delta_{U-CoT+FS}$	-0.28	-0.05	4.81	1.78	-1.69	-1.66	-0.67	-1.77	1.25	1.25	0.44	0.26
	LLaVa + Qwen	U-CoT	69.10	68.83	–	–	–	–	–	–	–	–	–	–
		U-CoT+	72.00	71.98	–	–	–	–	–	–	–	–	–	–
		Δ_{U-CoT+}	2.95	3.18	–	–	–	–	–	–	–	–	–	–
		$\Delta_{U-CoT+FS}$	-1.13	-0.90	–	–	–	–	–	–	–	–	–	–
Mistral-12B	Qwen2VL-7B	U-CoT	68.30	68.09	65.54	63.48	58.87	58.32	65.10	62.53	73.00	72.99	62.72	62.16
		U-CoT+	71.20	71.19	83.90	82.41	63.10	62.01	59.06	57.33	73.70	73.36	69.03	68.07
		Δ_{U-CoT+}	3.98	4.18	14.76	14.94	4.51	4.04	-3.02	-2.74	2.28	2.16	6.46	6.49
		$\Delta_{U-CoT+FS}$	0.33	0.17	1.20	-4.78	1.77	-1.74	-0.33	-3.22	1.15	0.77	0.79	-1.41
	LLaVa1.6-7B	U-CoT	65.30	64.60	62.43	54.60	58.03	56.53	61.74	56.89	76.40	76.38	62.33	61.68
		U-CoT+	70.00	69.88	81.07	78.57	59.72	56.40	64.43	61.36	75.60	75.33	68.05	67.82
		Δ_{U-CoT+}	4.60	5.20	16.95	21.59	1.41	-0.26	2.69	4.43	0.82	0.77	5.82	6.33
		$\Delta_{U-CoT+FS}$	-0.30	-0.52	4.24	-1.23	-1.20	-6.24	-0.67	-5.70	-1.88	-2.14	2.17	0.65
	LLaVa + Qwen	U-CoT	67.20	66.70	–	–	–	–	–	–	–	–	–	–
		U-CoT+	72.90	72.87	–	–	–	–	–	–	–	–	–	–
		Δ_{U-CoT+}	4.22	4.70	–	–	–	–	–	–	–	–	–	–
		$\Delta_{U-CoT+FS}$	-0.73	-0.96	–	–	–	–	–	–	–	–	–	–
Llama3.1-8Bf	Qwen2VL-7B	U-CoT+	68.00	66.98	71.19	61.44	61.13	56.21	68.46	62.35	71.90	71.86	63.51	60.86
		Δ_{U-CoT+}	4.50	5.54	8.76	10.80	-1.12	-2.57	4.70	4.80	2.10	2.77	2.17	2.96
	LLaVa1.6-7B	U-CoT+	67.00	65.51	72.03	60.88	57.75	51.09	63.76	53.86	73.50	73.50	64.50	63.26
		Δ_{U-CoT+}	4.00	5.15	6.78	14.22	-1.97	-5.31	0.67	2.02	4.80	5.40	3.95	4.77
	LLaVa+Qwen	U-CoT+	65.70	64.10	–	–	–	–	–	–	–	–	–	–
		Δ_{U-CoT+}	2.90	3.95	–	–	–	–	–	–	–	–	–	–
Qwen2.5-72Bf	Qwen2VL-7B	U-CoT+	67.50	67.50	70.34	69.97	62.54	60.20	62.42	62.13	73.60	73.39	67.26	66.90
		Δ_{U-CoT+}	0.80	1.05	21.75	22.90	3.95	2.74	5.37	5.32	1.80	1.59	3.75	4.52
	LLaVa1.6-7B	U-CoT+	65.70	65.53	77.12	75.26	59.15	54.99	67.11	66.63	74.40	74.40	64.30	64.00
		Δ_{U-CoT+}	3.00	3.59	17.23	15.44	1.12	-2.81	3.35	2.91	3.80	3.81	0.99	2.02
	LLaVa+Qwen	U-CoT+	66.60	66.55	–	–	–	–	–	–	–	–	–	–
		Δ_{U-CoT+}	2.80	3.32	–	–	–	–	–	–	–	–	–	–

Δ_{U-CoT+} : The average performance gain of U-CoT+ under guideline perturbations. $\Delta_{U-CoT+FS}$: The average performance gain of U-CoT+ with few-shot examples.

Our error analysis is conducted on a set of error cases with a balanced distribution of false negatives and false positives, randomly sampled from each of the three underperforming datasets (FHM, Harm-P, and PrideMM).

C FULL EXPERIMENT RESULTS

We present the detailed experiment results on all LMM+LLM combinations in Table 4, from which the results of the best-performing combination is highlighted in Table 1. We restrict our experiments to two 7B LMMs and do not extend to the larger LLaVa1.6-13B, as its M-CoT results do not consistently outperform the 7B counterpart across most datasets, with improvements observed only on Harm-P and MultiOFF, while incurring substantially higher computational overhead. Notably, the 13B model shares the same vision encoder and differs primarily in language model capacity, offering limited visual advantage. We therefore focus on 7B LMMs as a more efficient choice aligned with our low-resource design objective. By default, each unimodal LLM reasons over visual information extracted solely

by one paired 7B LMM. For FHM, we experiment with a specific collaborative setup between the two LMMs: Qwen2VL-7B is responsible for extracting disability-related visual cues, while LLaVa1.6-7B handles others. This design is motivated by Qwen2VL-7B’s lower tendency to hallucinate when identifying such information. This FHM-specific configuration is denoted as “LLaVa+Qwen” thereafter. In practice, practitioners can evaluate how well different LMMs’ (or their combinations’) visual understanding aligns with the target context based on sample memes to better determine which LMM setup offers better robustness in visual content recognition. Likewise, the selection of LLM+LMM combinations can be determined empirically according to pilot results.

As shown in Table 4, U-CoT+’s performance gains over U-CoT stay prominent across different LMM+LLM combinations in most datasets, indicating its overall effectiveness. Compared to more capable up-scaled LLMs, 7–8B LLMs struggle to achieve highly comparable results to that of state-of-the-art SFT baselines, and can

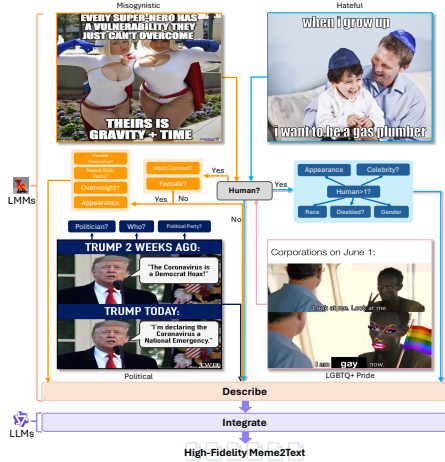
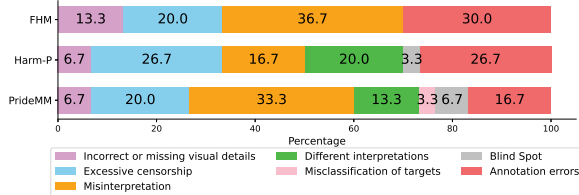
Table 5: Dataset Statistics.

Datasets	Test		Train
	#Harm	#Benign	
FHM [26]	490	510	8500
HarMeme [44]	124	230	3013
Harm-P [45]	171	184	2939
MultiOFF [57]	58	91	445
MAMI [17]	500	500	9000
PrideMM [52]	247	260	4328
GoatBench (Hateful) [31]	750	1250	–
GoatBench (Harmful)	420	589	–
GoatBench (Offensive)	303	440	–
GoatBench (Misogynistic)	500	500	–

Table 6: Prediction ensemble strategies. C.L.: Confidence Level for our human-crafted guidelines.

C.L.	Dataset	M-CoT	U-CoT	U-CoT+	Decision
–	ALL	–	–	1 0	Harmful
		–	–	0 1	
		–	–	1 1	
High	Harm-C, PrideMM	–	–	0 0	Harmless
Medium	FHM, MultiOFF	0 0	0 0	1 0	Harmless
		0 0	0 0	0 1	
		–	–	0 0	
Low	Harm-P, MAMI	0 0	0 0	–	Harmless
		–	–	0 0	

Note: 0/1: Predictions based on visual details extracted by Qwen2VL-7B.
 0/1: Predictions based on visual details extracted by LLaVA1.6-7B.

**Figure 5: Our proposed High-fidelity Meme2Text pipeline.****Figure 6: Error type distribution (%).**

only match or outperform GPT-4o-mini’s zero-shot results on FHM, HarMeme and MultiOFF. This suggests that, even when converted to a unimodal text-only inference setting, harmful meme detection remains reasoning-intensive, placing requirements on the general reasoning abilities of the underlying LLMs.

C.1 Guideline Perturbations

We apply three different perturbation strategies to our finalized human-crafted guidelines:

- **Rephrasing.** Each rule is reworded with GPT-4o to modify surface phrasing while largely maintaining the overall semantics.
- **Shuffling.** Both the order of rules and the pattern examples within rules are randomly shuffled.
- **Modifying.** The guidelines are altered by revising, adding, or removing rules, or by replacing with a GPT-4o-generated summary. *However, this strategy may compromise the coverage and integrity of the guidelines for some datasets.*

Table 4 shows that our method remains generally robust to these perturbations, though performance may occasionally fluctuate, particularly on challenging political meme datasets of MultiOFF and Harm-P, where models operate under high uncertainty, making them more sensitive to randomness and semantic nuances introduced by ambiguity from perturbations.

C.2 Guidelines vs. Few shots

We carefully select a small set of representative memes (10–20) from the holdout training split and use their GPT-4o-generated descriptions to construct a candidate pool of demonstration examples for each dataset. We evaluate four few-shot CoT settings with increasing numbers of shots of 4, 6, 8, and 10, where class-balanced examples are randomly sampled from the pool. Results show that positive performance gains $\Delta_{U-CoT+FS}$ can be found on some datasets, suggesting that our proposed unimodal harmful meme detection framework, built upon the Meme2Text pipeline, could be generalized to few-shot inference settings. Despite specific cases where the performance gap between guideline-based prompting and few-shot prompting is minimal (e.g., on MAMI and PrideMM), the overall in-context learning ability of lightweight LLMs remains limited in detecting harmful memes conveyed through text, making our human-crafted guidelines a more effective solution.

D LIMITATIONS AND FUTURE WORK

We consider the following limitations that we leave for future work: (i) We align our work with ongoing research developments and plan to extend our investigation to culturally diverse meme datasets (e.g., Mandarin, Hindi) as they become more widely available. (ii) Each application of our guideline-crafting procedure across the seven datasets serves as a test of U-CoT+’s cross-context transferability. We leave evaluation on more domain-specific datasets with diverse, underexplored types of harmful memes to future discussions. (iii) Our human-crafted guidelines offer reproducible, easy-to-follow templates that can shed light on future efforts toward automatic guideline generation using LLMs. (iv) Existing datasets lack detailed visual ground truth. Future work will manually assess the robustness of our Meme2Text pipeline to hallucination.